

Seventh Information Systems International Conference (ISICO 2023)

ChatGPT: The AI Game-Changing Revolution in Marketing Strategy for the Indonesian Cosmetic Industry

Wendra Wilendra^a, Reny Nadlifatin^{b*}, Cahya Khairani Kusumawulan^c

^a School of Interdisciplinary Management and Technology, Institut Teknologi Sepuluh Nopember, Surabaya, Indonesia

^b Department of Information Systems, Institut Teknologi Sepuluh Nopember, Kampus ITS Sukolilo, Surabaya, 60111, Indonesia

^c Department of Pharmacy, Faculty of Pharmacy, Padjadjaran University, Jl. Raya Bandung Sumedang KM 21, Jatinangor 45363, Indonesia;

Abstract

The Indonesian cosmetic industry has experienced rapid growth in recent years, requiring the adoption of innovative marketing strategies to remain competitive. This article explores the potential of ChatGPT, an advanced AI language model, in revolutionizing marketing strategy within the Indonesian cosmetic industry. Employing a qualitative research approach and thematic analysis, this study investigates AI-driven marketing strategies, product development, personalization in marketing, ethical considerations, and future research directions. The findings suggest that ChatGPT can significantly enhance customer targeting, market segmentation, pricing strategies, and promotion, as well as contribute to product development and personalized marketing efforts. However, ethical concerns and future research directions warrant further exploration in this rapidly evolving field.

© 2023 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<https://creativecommons.org/licenses/by-nc-nd/4.0>)

Peer-review under responsibility of the scientific committee of the Seventh Information Systems International Conference

Keywords: ChatGPT, Artificial Intelligence, Marketing Intelligence, Marketing Strategy, Indonesian Cosmetic Industry

1. Introduction

The Indonesian cosmetic industry has witnessed considerable growth in recent years, generating a revenue of US\$7.23 billion or IDR 111.83 trillion in 2022, with projections estimating an average annual growth rate of 5.81%

* Corresponding author. Tel.: +62 856-3002-176

E-mail address: reny@its.ac.id

from 2022 to 2027 [1][2]. This growth is attributed to evolving lifestyles, consumer behavior, and the rising middle class [3]. The number of cosmetic industries in Indonesia increased by 32% from 2021 to 2022, reaching 1,084 [4]. As a result, companies face challenges in adopting new technologies and adapting marketing strategies to meet changing market trends and consumer needs.

With growing competition and evolving technologies, marketing strategies in the cosmetic industry must become increasingly sophisticated. Companies now rely on more than traditional marketing techniques, embracing new technologies to expand their reach and enhance customer engagement. To address these challenges, companies must effectively analyze market data and trends, utilizing cutting-edge technologies like big data to gain deeper insights into their markets and consumers.

Big data presents significant opportunities for improving marketing effectiveness and efficiency. However, it also poses challenges that require companies to manage their data effectively, enhance data analysis skills, and improve data security. Research trends indicate that technologies like artificial intelligence (AI), machine learning, data visualization, and predictive models will continue to advance and shape future marketing strategies [5].

A Marketing Intelligence System (MIS) can aid companies in collecting, analyzing, and interpreting market data and trends to design effective marketing strategies. MIS systems play a crucial role in identifying new opportunities, threats, and providing accurate data for better understanding customer needs and making informed decisions [6].

AI, in particular, has become an essential tool for improving the efficiency and effectiveness of marketing activities. According to Raj Venkatesan and Jim Lecinski's "The AI Marketing Canvas: A Five-Stage Road Map to Implementing Artificial Intelligence in Marketing," AI can enhance data collection and analysis, predict customer behavior, and identify emerging market trends [7].

This study is intended to find the answer on how AI influences the cosmetic industry by exploring the potential of ChatGPT, an advanced AI language model developed by OpenAI [8], in revolutionizing marketing intelligence and strategy within the Indonesian cosmetic industry.

2. Literature Review

2.1. The cosmetics industry in Indonesia

The Indonesian cosmetics industry has experienced rapid growth, becoming a significant contributor to the national economy and employing approximately 6 million people. Three key factors drive this industry's growth: a young population with an average age of 28 years, steady economic growth, and the influence of social media [3].

The industry covers the production, distribution, and marketing of various facial, body, and hair care products, including makeup, skincare, haircare, perfume, and personal hygiene items. It has witnessed rapid development in recent decades due to economic growth, changing consumer lifestyles, globalization, and increased internet penetration, which have made cosmetics products more accessible to a broader segment of society [9].

Innovation and research have also played a crucial role in the industry's development, leading to the discovery of new materials and formulas for more effective beauty care [10][11]. Several trends and issues shape the cosmetics industry, such as consumer awareness of product ingredients, sustainability and ethical production practices, and the role of technology in marketing and distribution.

Consumers are increasingly conscious of the health and environmental impacts of cosmetics products, seeking safe, natural, and eco-friendly alternatives. Moreover, the use of technology, including social media, e-commerce, and artificial intelligence, in marketing and distributing cosmetics products is a critical factor in determining a company's success in the industry [9][12][13].

2.2. Marketing Strategy in the Cosmetics Industry

The highly competitive cosmetics industry requires strong marketing strategies to succeed. Market analysis, segmentation, and target market identification are crucial to understanding consumer needs and preferences [14]. One of the key elements of a successful marketing strategy in the cosmetics industry is market analysis. This involves evaluating the size, growth, trends, and competitors in the market. Segmentation is also an important aspect of marketing in the cosmetics industry. This involves grouping consumers based on similar characteristics such as

demographics, lifestyle, needs, and preferences [15]. When selecting a target market, a company must consider factors such as growth potential, purchasing power, and consumer needs [6][15]. Brand positioning is also critical for a cosmetics company. Brand positioning differentiates a company from competitors and creates long-term value[16].

Effective communication channels and relevant content convey brand value to consumers, while efficient distribution and sales strategies ensure product availability and customer satisfaction. A marketing framework unifies information to optimize marketing goals and keep teams focused [17].

The marketing mix, first proposed by Borden in 1964 and refined by McCarthy in 1968, comprises four essential elements: Product, Price, Place, Promotion [18]. Over time, criticisms against the 4P marketing mix concept have emerged, leading some experts like Booms & Bitner (1981) to propose new models names 7P by adding Participants, Process, Physical Evidence [19]. Marketing mix models, alongside the STP (Segmentation, Targeting, Positioning) model [20], provide a basis for analyzing marketing in the cosmetics industry.

In addition to the 4P and 7P marketing mix models and the STP model, there are other modern marketing strategies that are relevant for analyzing marketing in the cosmetics industry. Modern strategies such as customer journey mapping, the "They Ask You Answer" (TAYA) approach, and the STEPPS framework further enhance marketing effectiveness. Customer journey mapping identifies opportunities for improvement [21], while the TAYA approach by Marcus Sheridan [22] addresses customer concerns through informative content. The STEPPS framework (Social Currency, Triggers, Emotion, Public, Practical Value, Stories) helps design viral marketing strategies [23]. By utilizing this framework, cosmetic companies can design marketing strategies that tap into these six factors to increase the likelihood of their products becoming viral and achieving a wider reach. By combining the traditional and modern framework, cosmetic companies can help companies in the cosmetics industry create effective marketing strategies that resonate with their target audience and ultimately drive sales and growth.

2.3. Role of AI and ChatGPT in Marketing Intelligence

Marketing Intelligence focuses on collecting, analyzing, and presenting market and competitor information to improve marketing strategy effectiveness [16][24]. In the cosmetics industry, it helps companies understand target markets, competitors, and growth opportunities by providing insights into consumer behavior, preferences, and trends as well as the performance of their own products and services compared to those of their competitors [24][25]. In addition, Marketing Intelligence can also be used in decision-making for marketing strategies. This decision-making system utilizes data and information gathered from marketing intelligence, allowing for more accurate and effective decision-making. System-based decision-making enables companies to make decisions based on accurate data and facts, reducing the risk of error in decision-making. Moreover, this system allows companies to make decisions quickly and accurately since the necessary data and information are readily available within the system [24] [25][26].

AI integration in marketing intelligence advances data analytics, customer segmentation, and personalization, allowing businesses to make informed decisions and better target customers [27][28][29]. AI technologies like machine learning, natural language processing, and computer vision are employed for market segmentation, targeting, strategy design, chatbots, and sentiment analysis. AI compiles data from various sources like social media, menus, online reviews, and websites, and uses it to generate and deliver relevant content to the target audience. The application has many benefits, including improving marketing operational efficiency, enhancing customer experience, and increasing business performance. AI can be used for data analysis, content personalization, and data-driven decision-making, as well as in programmatic advertising, chatbots, dynamic pricing, and retargeting. However, proper planning and supervision are required to ensure the security and compliance of AI implementation in marketing. Using AI in digital marketing can improve the effectiveness of marketing campaigns and provide customers with a better experience [7][27][28].

One of the rapidly evolving Artificial Intelligence (AI) is ChatGPT (Generative Pre-trained Transformer) developed by OpenAI. ChatGPT, a language generation model designed to produce text resembling human writing. The system is trained to understand and generate text in various languages, including Bahasa Indonesia. As an advanced language model, ChatGPT can assist in various aspects, including marketing strategies. ChatGPT is a technology that can function as a chatbot or virtual assistant that uses AI to understand and respond to human conversations. This technology is capable of mimicking human abilities in speaking and understanding natural language. ChatGPT uses deep learning algorithms, a machine learning method that uses a neural network architecture consisting of multiple layers. This neural network is trained using large data (big data) taken from various sources

such as books, articles, and human conversations. As a result, the system is able to mimic human abilities in speaking and understanding natural language, providing appropriate responses [8]. ChatGPT, as a state-of-the-art AI model, can analyze vast amounts of data and generate human-like responses, providing valuable insights for marketing professionals [8][28][30][31].

ChatGPT, an AI language generation model developed by OpenAI, can be used in marketing strategies for the cosmetic industry. It offers various benefits such as efficient and quick customer data analysis, enabling companies to design more effective marketing strategies and target specific market segments. ChatGPT can also be used as a customer support tool, handling queries and complaints automatically and reducing the workload of customer service teams. In addition to marketing, ChatGPT can be applied in various fields such as translation, chatbots, article writing, and virtual assistants. It can improve decision-making in marketing strategies by conducting market research, sentiment analysis, and responding to market demands quickly. The technology can help companies develop better marketing content, improve customer service, and respond to market demands more efficiently [31][32][33][34].

3. Methodology

This study employs a qualitative research approach and thematic analysis, as suggested by Denzin and Lincoln [35]. The thematic analysis methodology enables the identification, analysis, and reporting of patterns and themes within the data. The data sources for this study include journal, articles, books, and reports from various sources. In this study, we use thematic analysis to explore the role of AI and ChatGPT in the Indonesian cosmetic industry's marketing strategy, drawing on insights from a range of academic literature. This approach allows for a comprehensive understanding of the potential benefits, challenges, and future developments of AI and ChatGPT in marketing strategy. By employing a thematic analysis method, this study aims to explore the potential impact of ChatGPT on marketing strategies within the Indonesian cosmetic industry [35][36].

The process of conducting thematic analysis can be divided into six main steps, as outlined by Braun and Clarke [37]. First, become familiar with the data by thoroughly reading and immersing in its content. Second, generate initial codes to label and summarize interesting features in the data. Third, search for potential themes by identifying patterns and connections among the codes. Fourth, review and refine the themes to ensure their accuracy and coherence. Fifth, define and name each theme with concise and meaningful descriptions. Finally, produce a comprehensive report that describes the analysis process, presents the themes, and supports them with relevant examples.

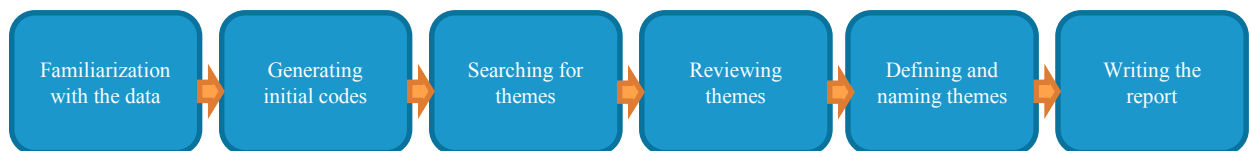


Fig. 1. Stages of thematic analysis

3.1. Data Collection

In order to develop a comprehensive database for this study, we identified relevant articles published between 2021 and 2023. We searched for papers using research databases such as Google Scholar, ScienceDirect, and Springer. Relevant articles were identified using key terms like "ChatGPT," "Artificial Intelligence," "Marketing Intelligence," and "Marketing Strategy," with the addition of "cosmetic" and/or "Indonesia" to specify the research field.

As a result, the specific search terms incorporated various combinations of these keywords, including "ChatGPT for Marketing," "Marketing Strategy Cosmetics", and "Artificial Intelligence for Marketing". This method allowed us to collect a wide-ranging and thorough set of sources to support our examination of potential ChatGPT applications in the marketing strategies of the Indonesian cosmetic industry.

4. Findings

4.1. Familiarization with the data

This involves reading and rereading the data to become familiar with its content and nuances. In this study, we reviewed academic literature, industry reports, and other relevant documents to gain an understanding of AI and ChatGPT's role in marketing strategy.

4.2. Generating initial codes

This step involves systematically coding the data to identify meaningful features and patterns. We used open coding to assign descriptive labels to relevant data segments, such as "customer targeting," "pricing strategies," and "ethical concerns".

Table 4.2. Results of Generating initial codes.

No.	Initial Code	Description
1	Customer Targeting	Use of AI and ChatGPT to identify and target specific customer segments
2	Market Segmentation	Analysis of market segments based on consumer preferences and behavior with the help of AI
3	Product Development	Utilizing ChatGPT to generate ideas and product suggestions based on market trends
4	Pricing Strategies	Optimal product pricing through AI analysis of market factor
5	Promotion and Distribution	ChatGPT recommendations for effective promotion and distribution strategies
6	Personalized Marketing	Creation of personalized marketing content based on consumer preferences
7	Ethical Concerns	Ethical concerns related to data privacy, algorithmic bias, and consumer manipulation
8	Future Research Directions	Identification of research areas that need further exploration

4.3. Searching for themes

In this stage, we reviewed the initial codes and grouped them into broader themes, which represent patterns or relationships within the data.

Table 4.3. Output of Searching for themes.

No.	Theme	Initial Codes Included
1	AI-driven Marketing Strategies	Customer Targeting, Market Segmentation, Pricing Strategies, Promotion and Distribution
2	ChatGPT in Product Development	Product Development
3	Personalization in Marketing	Personalized Marketing
4	Ethical and Social Considerations	Ethical Concerns
5	Future Research Directions	Future Research Directions

4.4. Reviewing themes

This step entails refining the themes to ensure they accurately represent the data. We reviewed our themes in relation to the coded data segments and the broader dataset, making adjustments as necessary to ensure a coherent and consistent analysis.

4.5. Defining and naming themes

In this stage, we developed clear definitions and names for each theme, capturing their essence and scope. For example, the theme "marketing strategy" was defined as "the application of AI and ChatGPT in various aspects of marketing, such as market segmentation, product development, pricing, promotion, and distribution."

Table 4.5. Findings from Defining and naming themes.

No.	Theme	Theme Description
1	AI-driven Marketing Strategies	This theme explores how AI and ChatGPT are utilized in customer targeting, market segmentation, pricing strategies, and promotion and distribution within the Indonesian cosmetic industry.
2	ChatGPT in Product Development	This theme focuses on the role of ChatGPT in generating ideas and suggestions for new products based on market trends and consumer preferences.
3	Personalization in Marketing	This theme discusses the role of ChatGPT in creating personalized marketing content tailored to individual consumer preferences and how it can enhance customer engagement and retention.
4	Ethical and Social Considerations	This theme addresses the ethical concerns related to data privacy, algorithmic bias, and consumer manipulation that arise from the use of AI and ChatGPT in marketing strategies.
5	Future Research Directions	This theme identifies areas of research that need further exploration in the context of AI, ChatGPT, and their applications in the Indonesian cosmetic industry's marketing strategies.

By examining these themes, we were able to gain a deeper understanding of the potential benefits, challenges, and future developments of AI and ChatGPT in the Indonesian cosmetic industry's marketing strategy.

4.6. Writing the report

Our thematic analysis of the available literature revealed several key themes related to the application of AI and ChatGPT in the Indonesian cosmetic industry's marketing strategy:

- Marketing intelligence: AI and ChatGPT enable more efficient and accurate data analysis, leading to better-informed marketing decisions and strategies [31][38].
- Customer targeting and segmentation: AI-driven analysis of consumer preferences and behavior helps businesses identify and target specific customer segments with tailored marketing messages [14][27][39][40][41][42].
- Product development: ChatGPT can generate innovative product ideas and suggestions based on market trends and consumer preferences [39][27][32][43][30][31][44][34].
- Pricing strategies: AI can help optimize pricing strategies by analyzing factors such as competitors' prices, product demand, and market conditions [39][27][32][43][30][31][44][34].
- Promotion and distribution: ChatGPT can recommend the most effective promotion and distribution channels based on consumer preferences and market trends [39][42] [14].
- Personalized marketing: AI and ChatGPT can facilitate the creation of personalized marketing content, including tailored product recommendations and promotions, to enhance customer engagement and satisfaction [39][27][32][43][30][31][44][34].
- Ethical concerns: The use of AI and ChatGPT in marketing raises ethical concerns related to data privacy, algorithmic biases, and consumer manipulation [32][31].
- Future research directions: Our analysis identified several areas for future research, including the exploration of AI and ChatGPT's impact on customer satisfaction, brand loyalty, and long-term business performance, as well as the development of ethical guidelines for AI-driven marketing [27][7] [26].

The final step involves integrating the thematic analysis into a coherent and compelling narrative. In this study, we presented the findings of our analysis in a structured format, discussing the role of AI and ChatGPT in marketing strategy, its potential benefits and challenges, and future research directions.

5. Conclusion

The integration of ChatGPT into marketing strategies has the potential to revolutionize the Indonesian cosmetic industry. First, AI and ChatGPT enable more efficient and accurate data analysis, leading to better-informed marketing decisions and strategies. Second, AI-driven analysis of consumer preferences and behavior aids in customer targeting and segmentation, allowing businesses to tailor marketing messages to specific segments. Third, ChatGPT can contribute to product development by generating innovative ideas based on market trends and consumer preferences. Fourth, AI assists in optimizing pricing strategies by analyzing competitors' prices, product demand, and market conditions. Fifth, ChatGPT can recommend effective promotion and distribution channels, enhancing the overall marketing mix. Sixth, AI and ChatGPT facilitate personalized marketing by creating tailored content, including product recommendations and promotions, to enhance customer engagement and satisfaction.

6. Future Research Directions

This study provides an overview of the potential benefits of ChatGPT in the Indonesian cosmetic industry's marketing strategy. However, the use of AI and ChatGPT in marketing raises ethical concerns related to data privacy, algorithmic biases, and consumer manipulation. Future research can explore on exploring the impact of AI and ChatGPT on customer satisfaction, measuring the impact on customer engagement, brand loyalty, brand perception, sales performance, and long-term business performance, as well as the development of ethical guidelines for AI-driven marketing. Lastly, researchers can explore other AI technologies and their potential application in the Indonesian cosmetic industry, providing a broader perspective on the role of AI in marketing strategy.

References

- [1] KEMENPERIN RI. Kemenperin: Industri Kosmetik Nasional Tumbuh 20%. KemenperinGoId 2018. <https://www.kemenperin.go.id/artikel/18957/Industri-Kosmetik-Nasional-Tumbuh-20> (accessed March 14, 2023).
- [2] Statista. Beauty & Personal Care - Indonesia | Market Forecast. StatistaCom 2023. <https://www.statista.com/outlook/cmo/beauty-personal-care/indonesia> (accessed March 14, 2023).
- [3] KEMENPERIN RI. Kemenperin: Perubahan Gaya Hidup Dorong Industri Kosmetik. KemenperinGoId 2020. <https://kemenperin.go.id/artikel/21460/Perubahan-Gaya-Hidup-Dorong-Industri-Kosmetik> (accessed March 15, 2023).
- [4] BPOM RI. Portal Satu Data BPOM. SatudataPomGoId 2023. <https://satudata.pom.go.id/datasets/14-jumlah-industri-kosmetik-di-indonesia?lang=id&tabs=tabs1> (accessed March 14, 2023).
- [5] SA M, Rahim A, Abughazaleh ZM. Big Data in Marketing Arena: Big Opportunity/Benefit, Big Challenge, and Research Trends: An Integrated View. *International Journal of Economics & Management Sciences* 2018;07. <https://doi.org/10.4172/2162-6359.1000533>.
- [6] Kotler & Keller. *Marketing Management*. 2016.
- [7] Venkatesan R, Lecinski J. *The AI marketing canvas : a five-stage road map to implementing artificial intelligence in marketing*. vol. 1. STANFORD BUSINESS BOOKS; 2021.
- [8] OpenAI. GPT-4 is OpenAI's most advanced system, producing safer and more useful responses 2023.
- [9] Nielsen. Indonesia FMCG Growth hit 11% in 2018, Driven by Food and Beverage Spending 2018.
- [10] Subriadi AP, Baturohmah H. Social media in marketing of ride-hailing: A systematic literature review. *Procedia Computer Science* 2021;197:102–9. <https://doi.org/10.1016/j.procs.2021.12.123>.
- [11] Mudjahidin, Sholichah NL, Aristio AP, Junaedi L, Saputra YA, Wiratno SE. Purchase intention through search engine marketing: E-marketplace provider in Indonesia. *Procedia Computer Science* 2021;197:445–52. <https://doi.org/10.1016/j.procs.2021.12.160>.
- [12] globaldata. COVID-19 Consumer Survey: Comfort, Uncertainty and Indulgence - Impact on consumer behavior in Asia-Pacific - Week 8 2021.
- [13] Euromonitor. Beauty and Personal Care in Indonesia | Market Research Report | Euromonitor 2021. <https://www.euromonitor.com/beauty-and-personal-care-in-indonesia/report> (accessed March 15, 2023).
- [14] Shaw EH. Marketing strategy: From the origin of the concept to the development of a conceptual framework. *Journal of Historical Research in Marketing* 2012;4:30–55. <https://doi.org/10.1108/17557501211195055>.
- [15] Kotler P, Armstrong G. *Principles of Marketing*. 2018.
- [16] Hedin H, Hirvensalo I, Vaarnas M. *The Handbook of Market Intelligence*. John Wiley & Sons, Ltd; 2016.

- [17] AMA. What Are Marketing Frameworks? 2022.
- [18] Boarden HN. The Concept of the Marketing Mix. *Journal of Advertising Research* 1964.
- [19] Booms, B. H. & Bitner MJ. Marketing strategies and organization structures for service firms. American Marketing Association 1981.
- [20] Goworek H, McGoldrick P. *Retail Marketing Management: Principles and Practice*, Pearson, UK. 2015.
- [21] Hamilton R, Price LL. Consumer journeys: developing consumer-based strategy. *Journal of the Academy of Marketing Science* 2019;47:187–91. <https://doi.org/10.1007/s11747-019-00636-y>.
- [22] Sheridan M. They Ask, You Answer: A Revolutionary Approach to Inbound Sales, Content - Marcus Sheridan - Google Books. 2019.
- [23] Patrik L. Social Media Marketing as Tool for a Small Company-Case: FinnEasy Oy 2022.
- [24] Ade LPK, Akanbi AM, Tubosun AI. The influence of marketing intelligence on business competitive advantage (a study of diamond bank Plc). *Journal of Competitiveness* 2017;9:51–71. <https://doi.org/10.7441/joc.2017.01.04>.
- [25] Eletter SF, Yasmin T, El Refae GA. Marketing intelligence in the era of big data. *TEM Journal* 2019;8:938–44. <https://doi.org/10.18421/TEM83-35>.
- [26] Alnaser AS. the Impact of Marketing Intelligence on Customer Brand Co-Creation of Geographical Indication Products: Case of Jordanian Mosaics. *Academy of Strategic Management Journal* 2021;20:1–12.
- [27] Haleem A, Javaid M, Asim Qadri M, Pratap Singh R, Suman R. Artificial intelligence (AI) applications for marketing: A literature-based study. *International Journal of Intelligent Networks* 2022;3:119–32. <https://doi.org/10.1016/j.ijin.2022.08.005>.
- [28] Verma S, Sharma R, Deb S, Maitra D. Artificial intelligence in marketing: Systematic review and future research direction. *International Journal of Information Management Data Insights* 2021;1:100002. <https://doi.org/10.1016/j.ijime.2020.100002>.
- [29] Dondapati A, Sheoliha N, Panduro-Ramirez J, Bakhare R, Sreejith PM, Devi Prasad Kotni V V. An integrated artificial intelligence framework for knowledge production and B2B marketing rational analysis for enhancing business performance. *Materials Today: Proceedings* 2022;56:2232–5. <https://doi.org/10.1016/j.matpr.2021.11.557>.
- [30] Shahriar S, Hayawi K. Let 's have a chat ! A Conversation with ChatGPT : Technology , Applications , and Limitations 2023.
- [31] Dwivedi YK, Kshetri N, Hughes L, Slade EL, Jeyaraj A, Kar AK, et al. “So what if ChatGPT wrote it?” Multidisciplinary perspectives on opportunities, challenges and implications of generative conversational AI for research, practice and policy. *International Journal of Information Management* 2023;71. <https://doi.org/10.1016/j.ijinfomgt.2023.102642>.
- [32] Lim WM, Gunasekara A, Pallant JL, Pallant JI, Pechenkina E. Generative AI and the future of education: Ragnarök or reformation? A paradoxical perspective from management educators. *International Journal of Management Education* 2023;21:100790. <https://doi.org/10.1016/j.ijme.2023.100790>.
- [33] Cotton DRE, Cotton PA, Shipway JR. Chatting and Cheating: Ensuring academic integrity in the era of ChatGPT. ... <https://doi.org/10.1016/j.osf.2023.103554>.
- [34] Graf A, Bernardi RE. ChatGPT in Research: Balancing Ethics, Transparency and Advancement. *Neuroscience* 2023;515:71–3. <https://doi.org/10.1016/j.neuroscience.2023.02.008>.
- [35] Norman K. Denzin, Yvonne Lincoln and YSL. *The SAGE handbook of qualitative research*. 2005.
- [36] Creswell JW, Poth CN. *Qualitative inquiry & research design : choosing among five approaches*. 2018.
- [37] Braun V, Clarke V. Using thematic analysis in psychology. *Qualitative Research in Psychology* 2006;3:77–101. <https://doi.org/10.1191/1478088706qp0630a>.
- [38] Haleem A, Javaid M, Pratap R. BenchCouncil Transactions on Benchmarks , Standards and Evaluations An era of ChatGPT as a significant futuristic support tool : A study on features , abilities , and challenges. *BenchCouncil Transactions on Benchmarks, Standards and Evaluations* 2023;2:100089. <https://doi.org/10.1016/j.tbench.2023.100089>.
- [39] Dumitriu D, Popescu MAM. Artificial intelligence solutions for digital marketing. *Procedia Manufacturing* 2020;46:630–6. <https://doi.org/10.1016/j.promfg.2020.03.090>.
- [40] Xiao-Yuan LIU. Agricultural products intelligent marketing technology innovation in big data era. *Procedia Computer Science* 2021;183:648–54. <https://doi.org/10.1016/j.procs.2021.02.110>.
- [41] Oktay S. An analytical study to identify and determine the usage frequency of sales and marketing strategies for 5 star hotels in the Antalya region. *Procedia Computer Science* 2017;120:862–70. <https://doi.org/10.1016/j.procs.2017.11.319>.
- [42] Halim KK, Halim S, Felecia. Business intelligence for designing restaurant marketing strategy: A case study. *Procedia Computer Science* 2019;161:615–22. <https://doi.org/10.1016/j.procs.2019.11.164>.
- [43] Chatterjee L, Feng C, Nakata C, Sivakumar K. The environmental turbulence concept in marketing: A look back and a look ahead. *Journal of Business Research* 2023;161:113775. <https://doi.org/10.1016/j.jbusres.2023.113775>.
- [44] Choi EPH, Lee JJ, Ho MH, Kwok JYY, Lok KYW. Chatting or cheating? The impacts of ChatGPT and other artificial intelligence language models on nurse education. *Nurse Education Today* 2023;125:714919. <https://doi.org/10.1016/j.nedt.2023.105796>.